



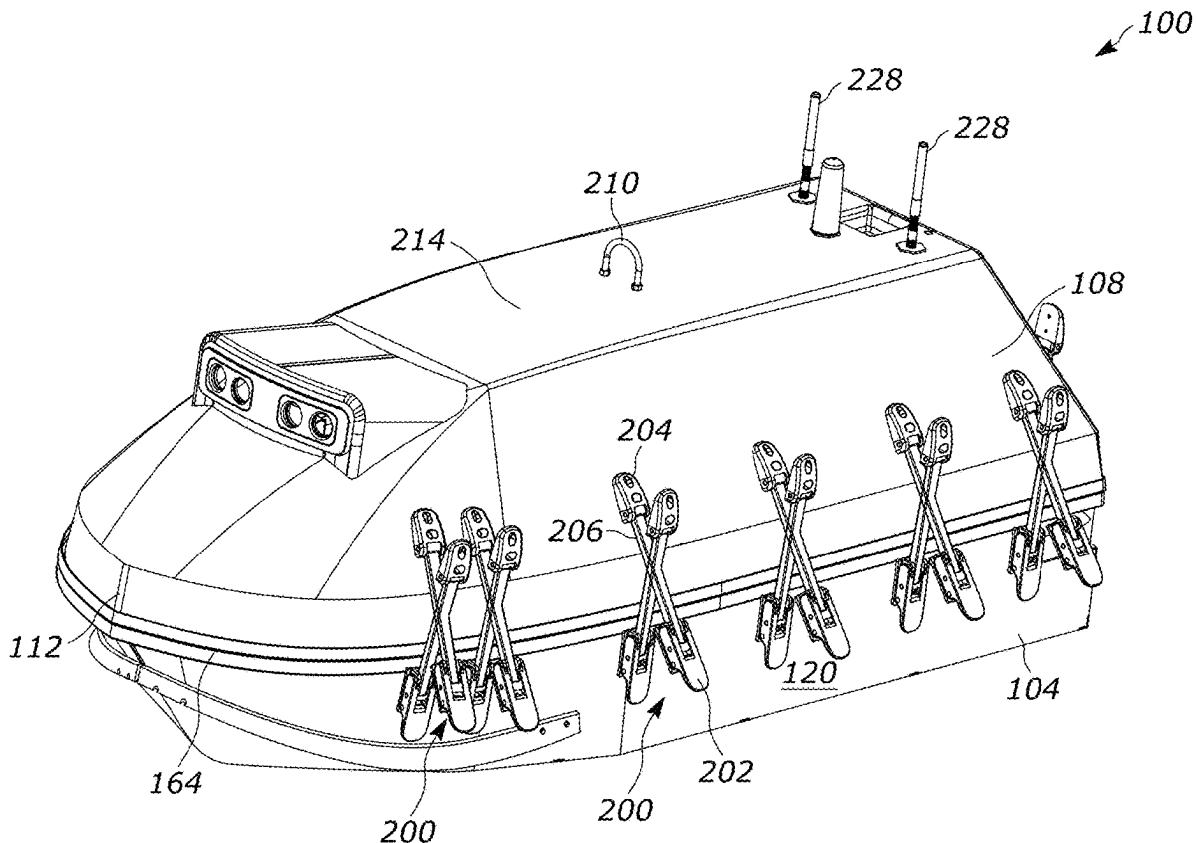
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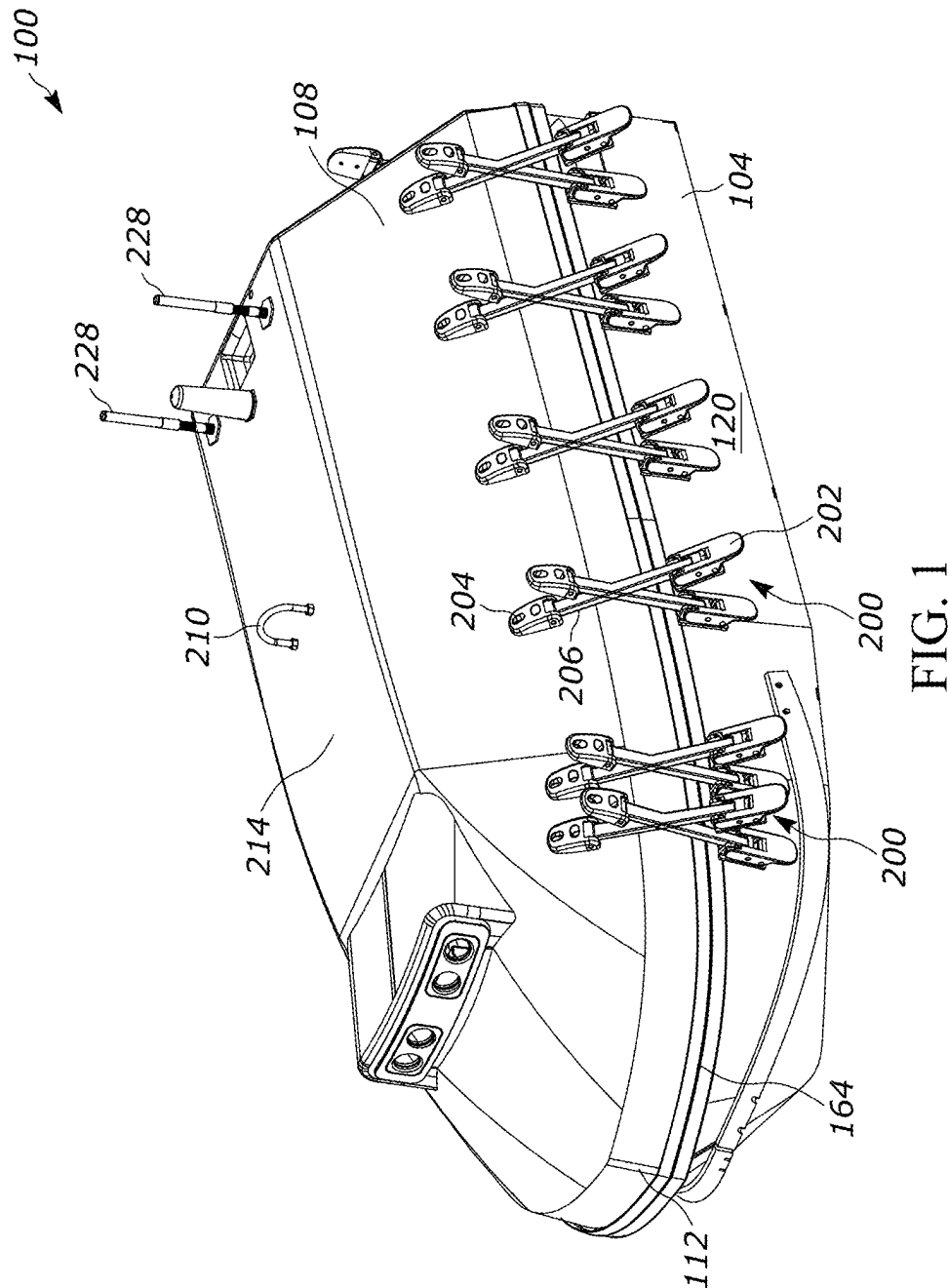
(19) **United States**(12) **Patent Application Publication****Howard et al.**(10) **Pub. No.: US 2025/0276770 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **MARITIME VEHICLE WITH REMOVABLE
CAP HAVING A PICK POINT****Publication Classification**(51) **Int. Cl.****B63B 35/00** (2020.01)**B63B 19/00** (2006.01)(52) **U.S. Cl.**CPC **B63B 35/00** (2013.01); **B63B 2019/0069**
(2013.01); **B63B 2035/007** (2013.01)(71) Applicant: **SARONIC TECHNOLOGIES**, Austin,
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Kubicki**, Austin, TX (US)(21) Appl. No.: **19/070,351**(22) Filed: **Mar. 4, 2025****Related U.S. Application Data**(60) Provisional application No. 63/561,301, filed on Mar.
4, 2024, provisional application No. 63/561,166, filed
on Mar. 4, 2024, provisional application No. 63/561,
063, filed on Mar. 4, 2024, provisional application
No. 63/561,282, filed on Mar. 4, 2024, provisional
application No. 63/561,181, filed on Mar. 4, 2024,
provisional application No. 63/561,102, filed on Mar.
4, 2024.

(57)

ABSTRACT

An autonomous maritime surface vehicle (AMSV) that includes a hull and a cap removably coupled to the hull (e.g., via a plurality of latches). The cap is movable between a first configuration, in which the cap is coupled to and sealingly encloses an entirety of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting access to the interior of the hull. The AMSV also includes a single pick point coupled to a top, exterior surface of the cap, such that the AMSV can be easily and efficiently picked up and carried via the single pick point.





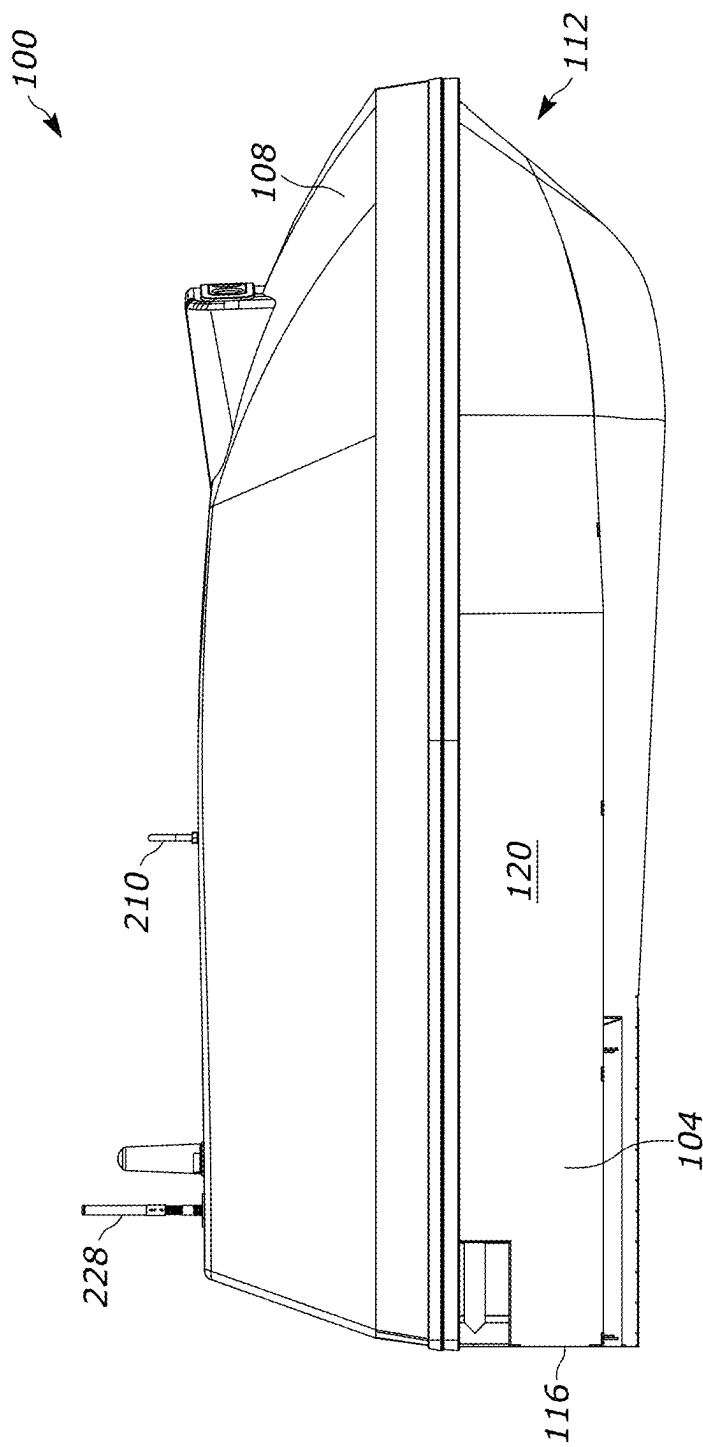


FIG. 2

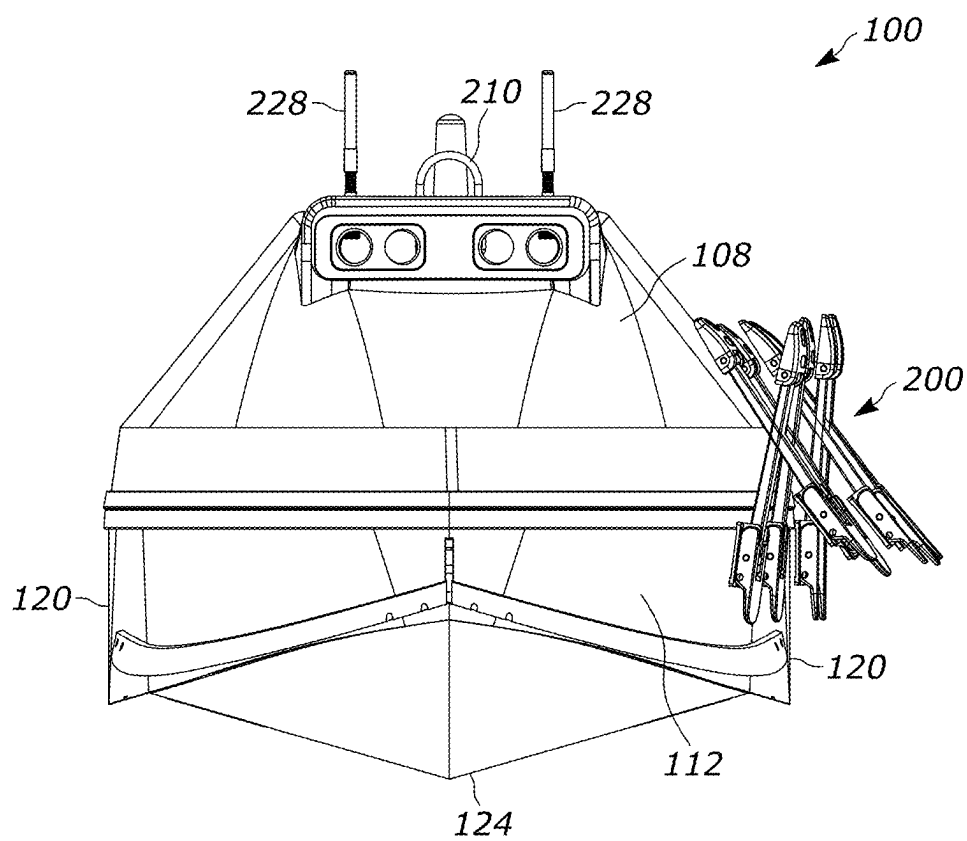


FIG. 3

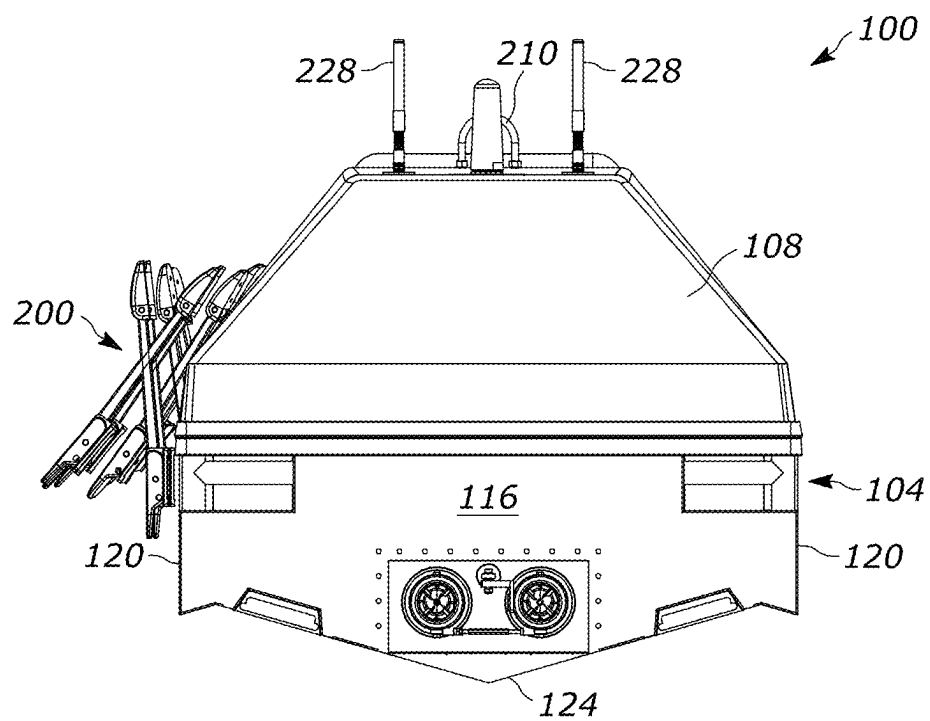


FIG. 4

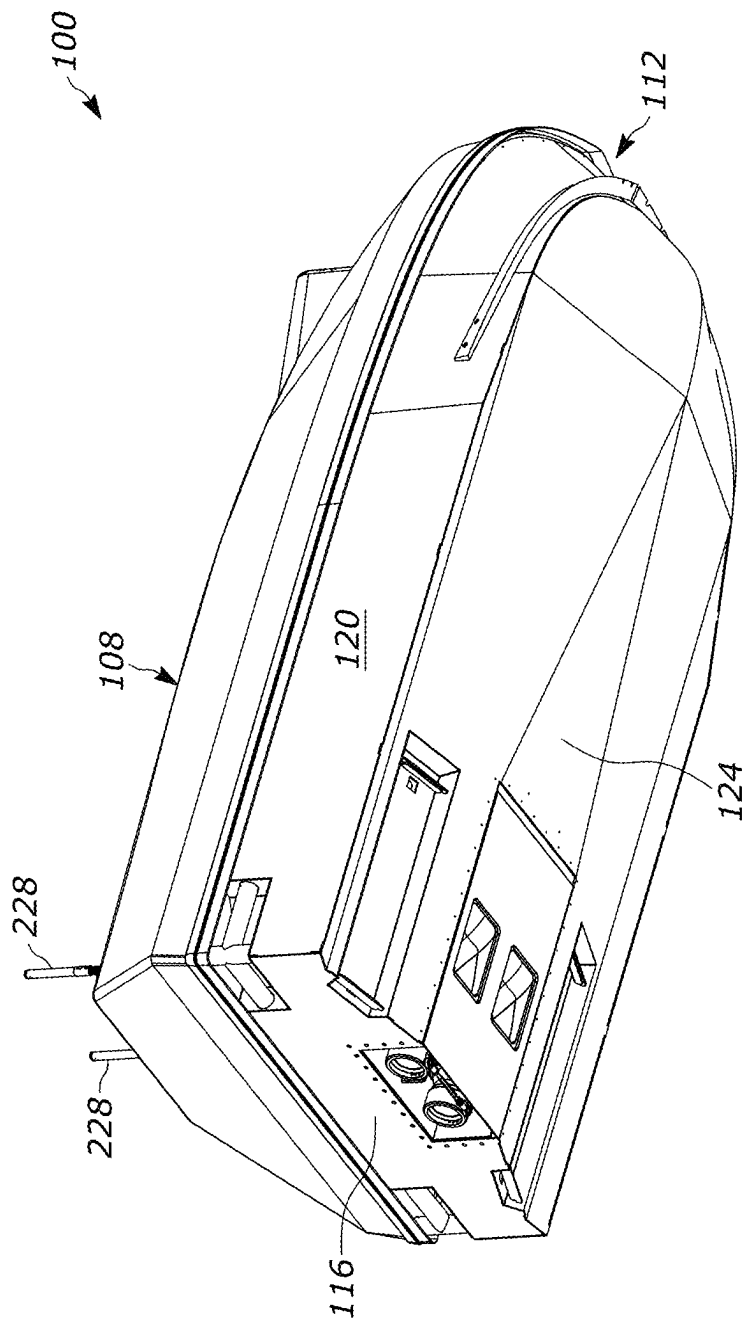


FIG. 5

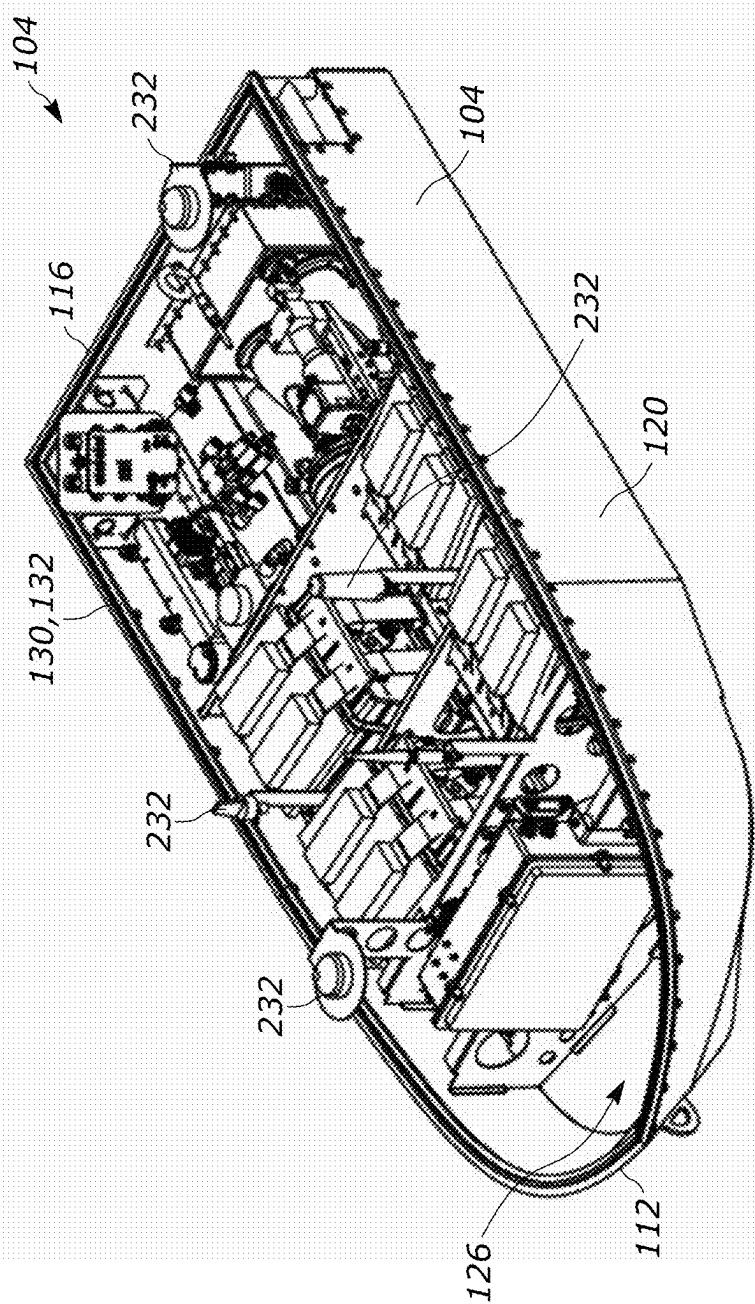


FIG. 6A

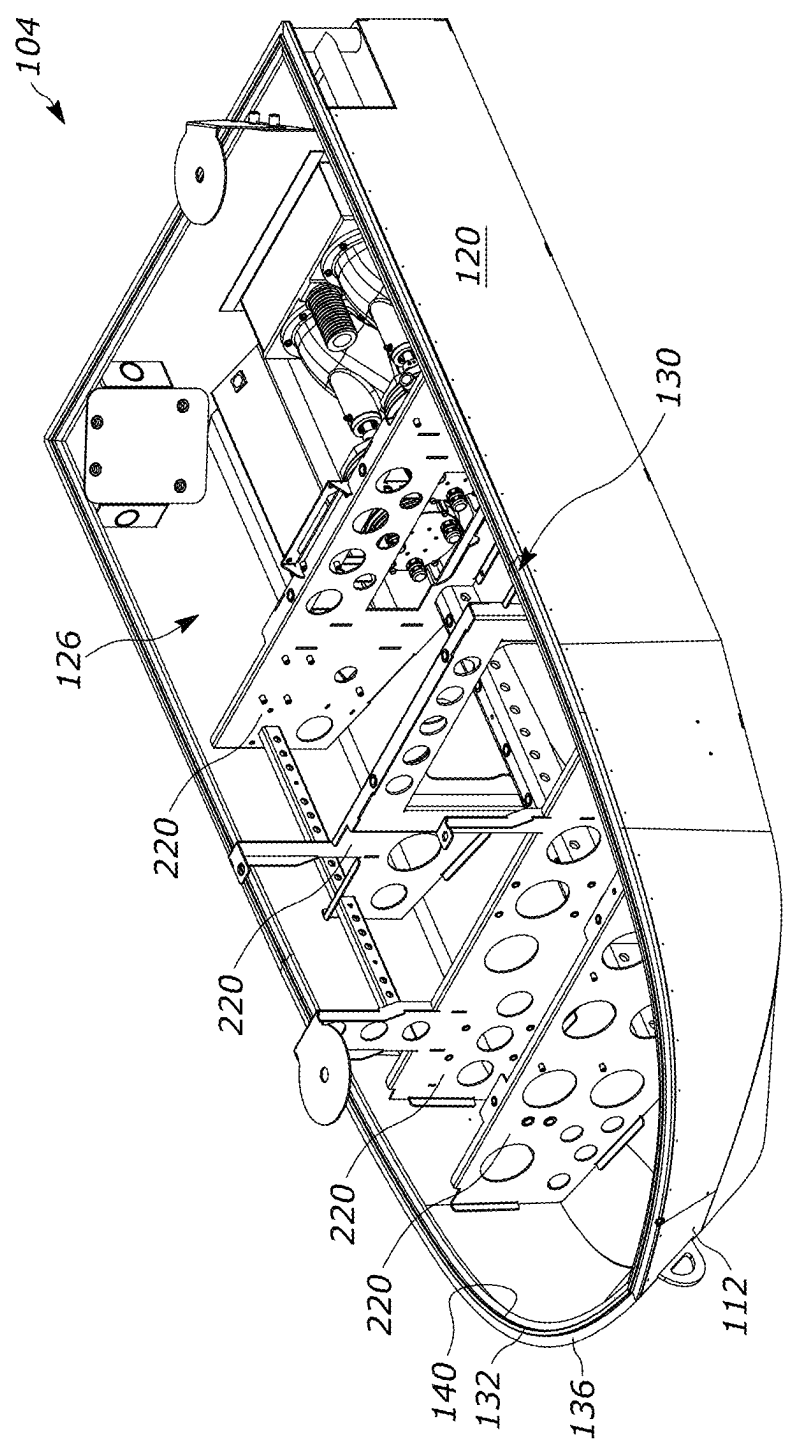


FIG. 6B

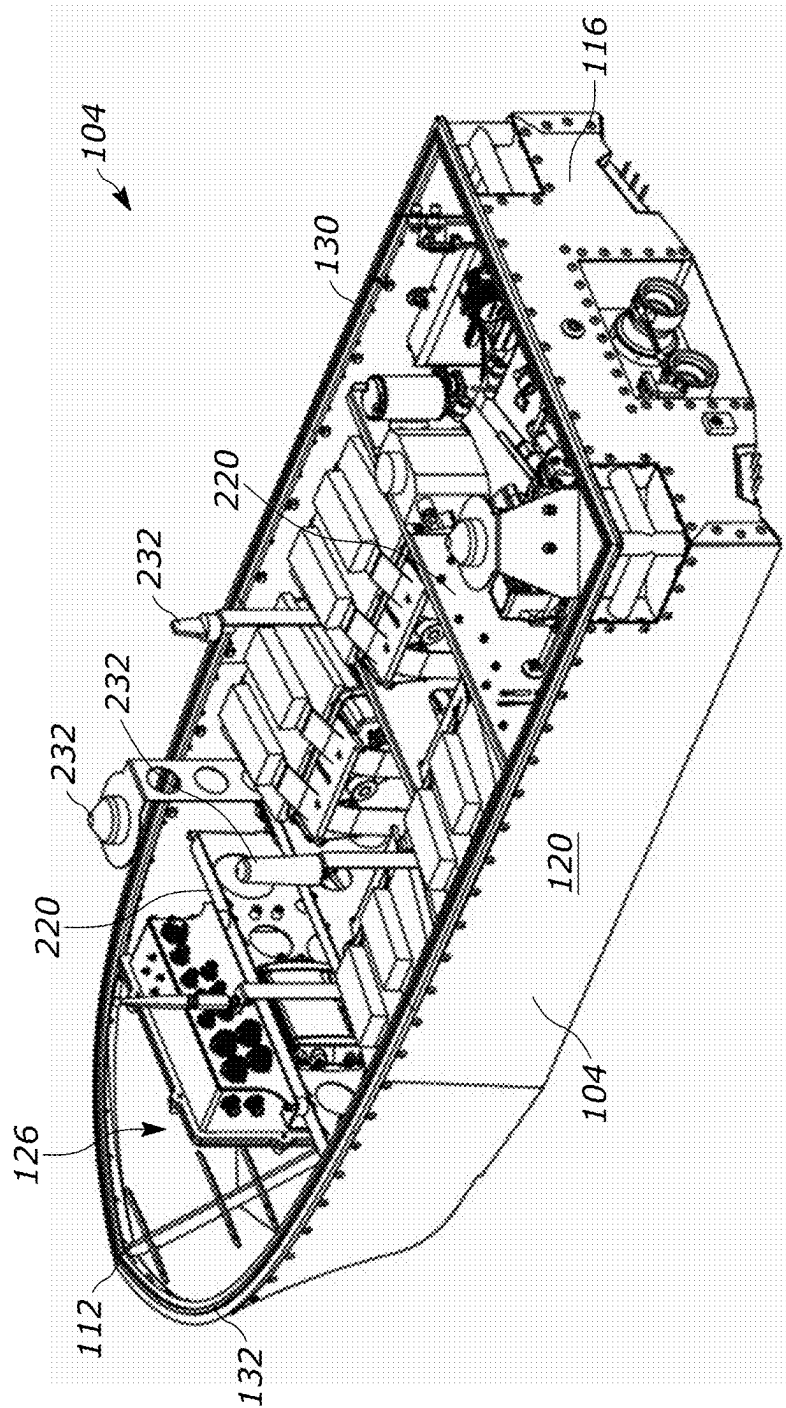


FIG. 6C

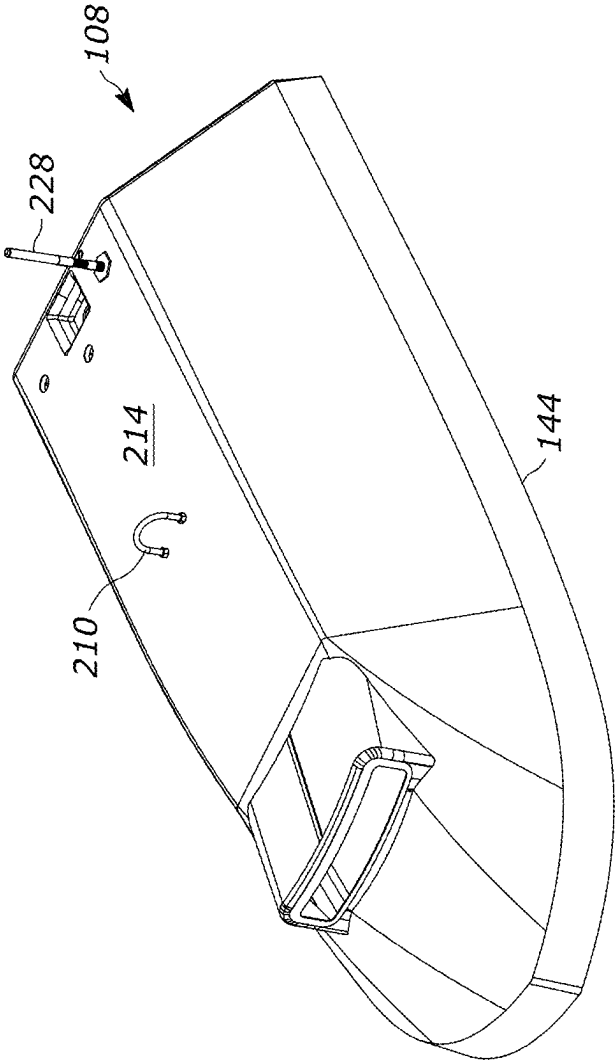


FIG. 7A

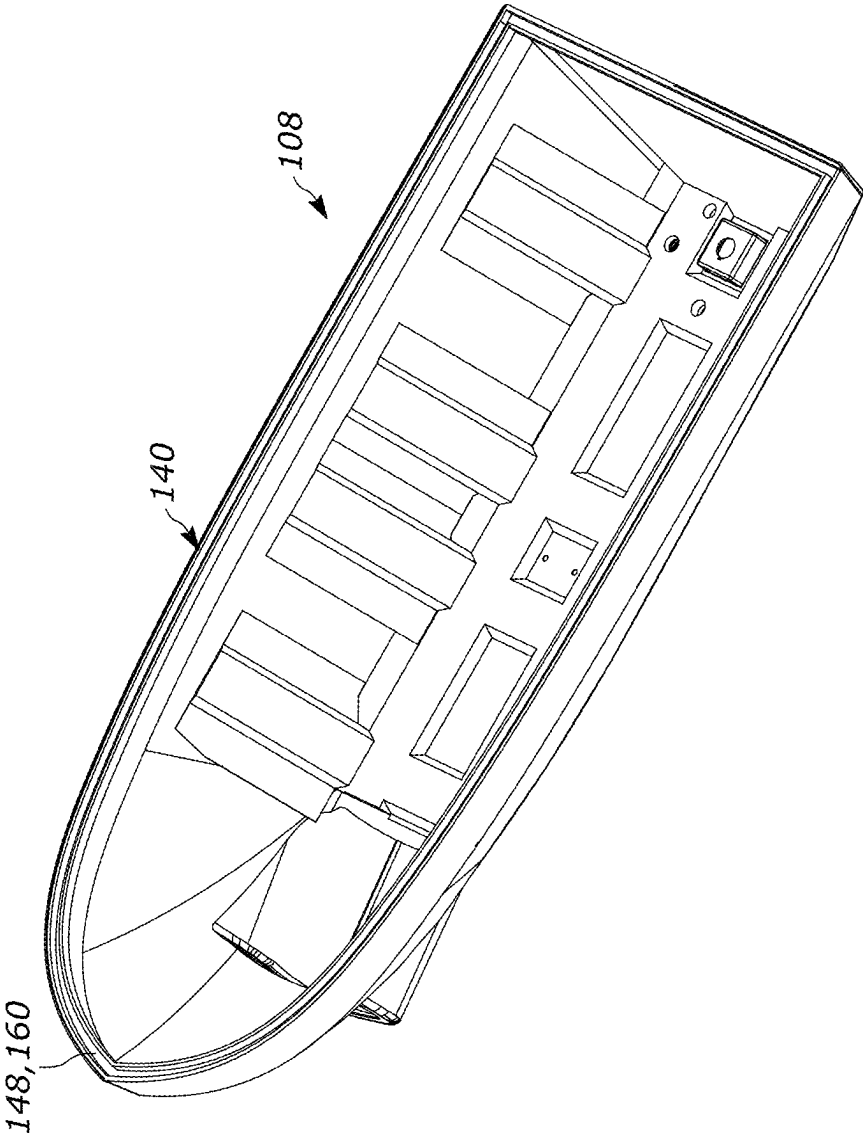


FIG. 7B

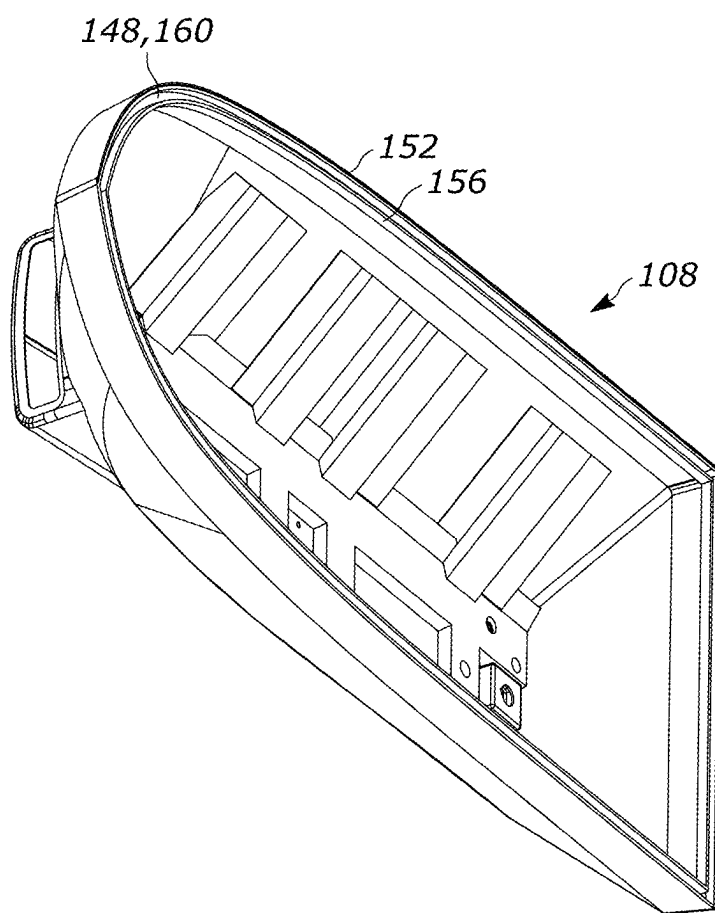


FIG. 7C

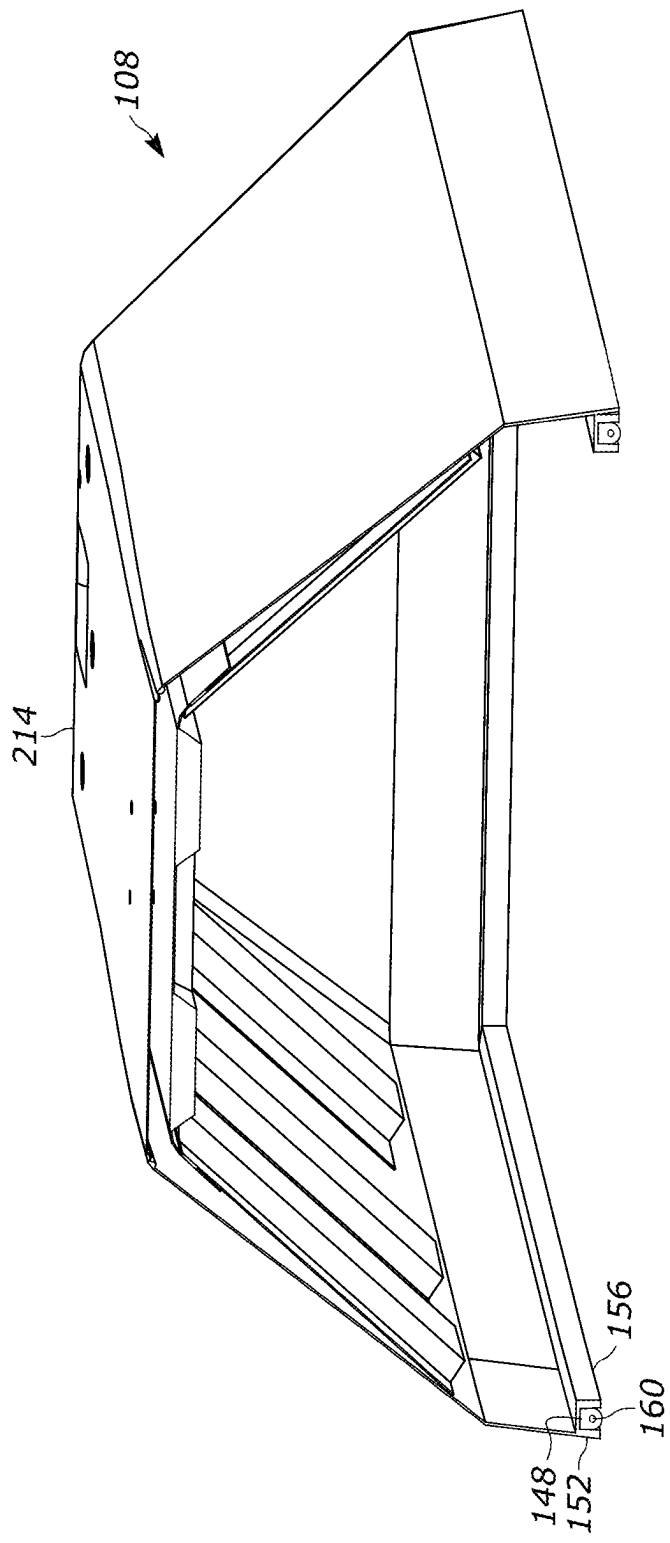


FIG. 7D

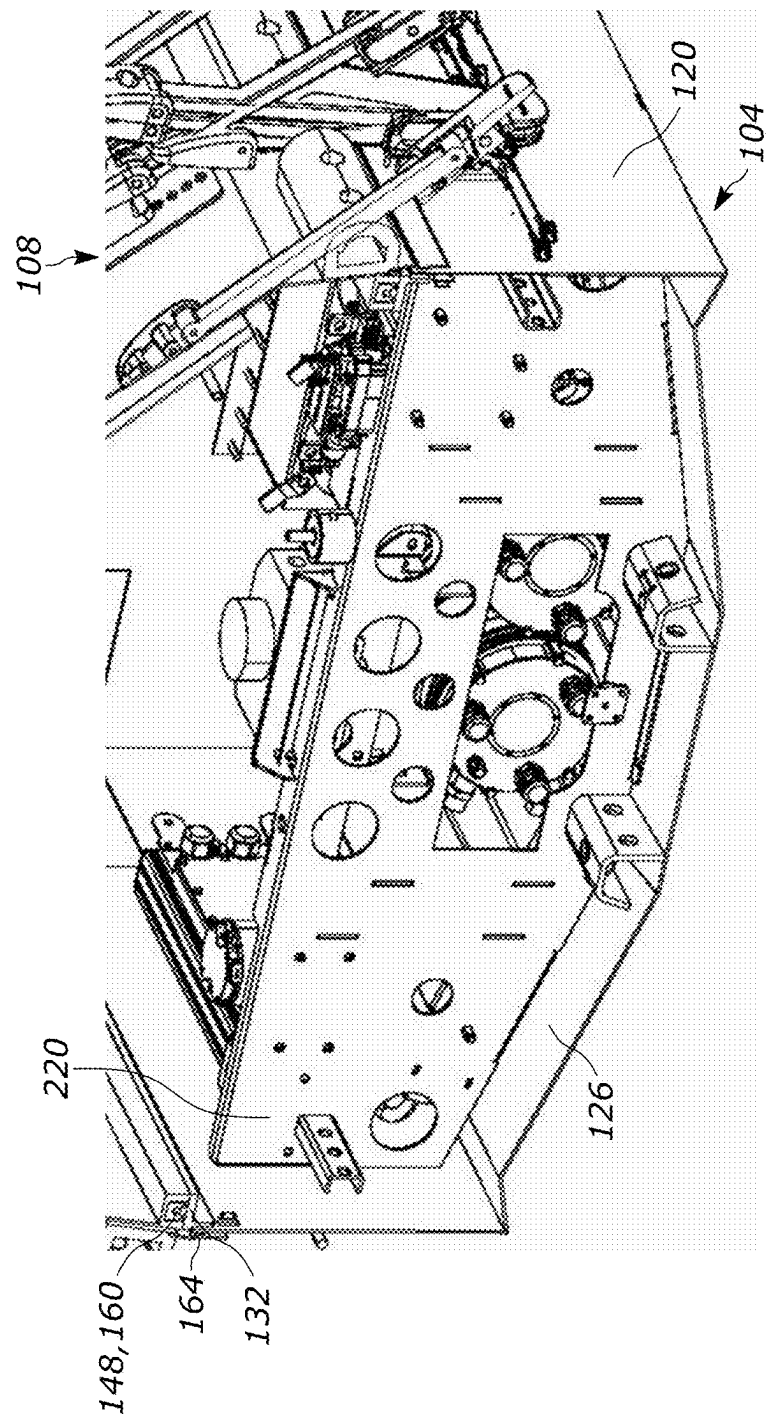


FIG. 8

MARITIME VEHICLE WITH REMOVABLE CAP HAVING A PICK POINT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to each of the following: U.S. Provisional Patent Application No. 63/561,301, titled “Vision System for Maritime Vehicle,” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,166, titled “Jet Pump Assembly for Maritime Vehicle” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,063, titled “Power System for Maritime Vehicle” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,282, titled “Systems and Approaches for Assembling a Maritime Vehicle” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,181, titled “Micro-Keel Cooler for Maritime Vehicle” and filed on Mar. 4, 2024; and U.S. Provisional Patent Application No. 63/561,102, titled “Maritime Vehicle with Recessed Tow Points” and filed on Mar. 4, 2024. The contents of each of these disclosures is hereby incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to maritime vehicles and more specifically to a maritime vehicle with a removable cap having a pick point.

BACKGROUND OF THE DISCLOSURE

[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.

SUMMARY

[0004] An autonomous maritime surface vehicle (AMSV) that includes a hull and a cap removably coupled to the hull (e.g., via a plurality of latches). The cap is movable between a first configuration, in which the cap is coupled to and sealingly encloses an entirety of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting access to the interior of the hull. The AMSV also includes one or more pick points coupled to a top, exterior surface of the cap, such that the AMSV can be easily and efficiently picked up and carried via the one or more pick points.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The features of this invention which are believed to be novel are set forth with particularity in the appended claims. The invention may be best understood by reference to the following description taken in conjunction with the accompanying drawings, in which like reference numerals identify like elements in the several FIGS., in which:

[0006] FIG. 1 is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure;

[0007] FIG. 2 is a side view of the maritime vehicle of FIG. 1;

[0008] FIG. 3 is a front view of the maritime vehicle of FIG. 1;

[0009] FIG. 4 is a rear view of the maritime vehicle of FIG. 1;

[0010] FIG. 5 is a bottom perspective view of the maritime vehicle of FIG. 1;

[0011] FIG. 6A is similar to FIG. 1, but with the cap and various components of the maritime vehicle removed for illustrative purposes;

[0012] FIG. 6B is similar to FIG. 6A, but with additional components of the maritime vehicle removed for illustrative purposes;

[0013] FIG. 6C is a rear, perspective view of the maritime vehicle of FIG. 6A;

[0014] FIG. 7A is a top, perspective view of the cap of the maritime vehicle of FIGS. 1-5;

[0015] FIG. 7B is a bottom, perspective view of the cap of FIG. 7A;

[0016] FIG. 7C is a close-up view of a portion of the cap of FIG. 7B;

[0017] FIG. 7D is a cross-sectional view taken along line D-D in FIG. 7A; and

[0018] FIG. 8 is a cross-sectional view taken along line 8-8 in FIG. 1.

DETAILED DESCRIPTION

[0019] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but can also be used for other purposes if desired. The maritime vehicle is small(er), durable, and configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.

[0020] FIGS. 1-8 illustrate one example of a maritime vehicle **100** constructed in accordance with the teachings of the present disclosure. The maritime vehicle **100** is an unmanned vessel configured to autonomously traverse a body of water. In other words, the maritime vehicle **100** is an autonomous maritime surface vessel (or vehicle). The maritime vehicle **100** generally includes a hull **104** and a cap **108** that is coupled to the hull **104** to secure various components within the maritime vehicle **100**. The hull **104** is at least partially disposed in the body of water in which the maritime vehicle **100** is traversing. The hull **104** in this example is a mono-hull that has a front (or bow) **112**, a rear (or stern) **116**, two sides **120**, and a keel **124** coupled to another. The front **112**, the rear **116**, the sides **120**, and the keel **124** can be welded together or can be coupled to one another in a different manner. For example, the front **112**, the rear **116**, the sides **120**, and the keel **124** can be coupled together in the manner described in U.S. Provisional Application No. 63/561,282, titled “Systems and Approaches for Assembling a Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference

herein. The hull 104 is configured such that the hull provides a continuous planning surface that allows the maritime vehicle 100 to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap 108 is coupled to the hull 104 to cover and/or conceal the components of the maritime vehicle 100 disposed in and carried by the hull 104 as the maritime vehicle 100 traverses the body of water.

[0021] In this example, the cap 108 is removably coupled to the hull 104 so as to form a clamshell configuration that selectively covers and/or conceal the components of the maritime vehicle 100 disposed in and carried by an interior 126 of the hull 104. More particularly, the cap 108 is reconfigurable, relative to the hull 104, between a first configuration or position (shown in FIGS. 1-4 and 8) and a second configuration or position (not explicitly illustrated). In the first configuration, the cap 108 is coupled to and sealingly encloses at least a portion of the interior 126 of the hull 104. In this example, when the cap 108 is in the first configuration, the cap 108 sealingly encloses an entirety of the interior 126 of the hull 104, such that the cap 108 entirely covers and conceals the components of the maritime vehicle 100 disposed in and carried by the hull 104. At the same time, the cap 108 substantially prevents any water from the body of water and objects in the water (or the ambient environment more generally) from entering the interior 126 of the hull 104 via the cap 108 (or an interface between the hull 104 and the cap 108). In other examples, however, the cap 108 may only sealingly enclose part of (and partially cover) the interior 126 when the cap 108 is in the first configuration. In the second configuration, the cap 108 is at least partially decoupled and spaced from the hull 104, thereby permitting access to the interior 126 of the hull 104. In some examples, the cap 108 will be completely decoupled, or removed, from the hull 104 in the second configuration (in which case the hull 104 will generally appear as shown in FIGS. 7A-7C). In other examples, however, the cap 108 may be partially coupled to and partially decoupled, or separated, from the hull 104 in the second configuration, in which case the cap 108 may still partially cover but still permit access to the interior 126 of the hull 104. For example, a first portion of the cap 108 may engage or be immediately adjacent to a first portion of the hull 104 while a second portion of the cap 108 is spaced from a second portion of the hull 104 (and the entire hull 104 more generally).

[0022] In this example, the hull 104 and the cap 108 each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can be equal to approximately 14 feet. The hull 104 is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle 100 can include two or more hulls (e.g., two parallel hulls) instead of the mono-hull.

[0023] As best illustrated in FIGS. 6A-6C, the hull 104 also includes a perimeter 130 and a projection 132 that extends around at least a portion of the perimeter 130. The perimeter 130 surrounds (and partially defines) the interior 126 and is generally defined by the front (or bow) 112, the rear (or stern) 116, and the two sides 120 of the hull 104. In this example, the projection 132 extends outward (upward in the orientation shown in FIGS. 6A-6C) from the front 112,

the rear 116, and the two sides 120 around an entirety of the perimeter 130. In other examples, however, the projection 132 may only extend around part of the perimeter 130 or the hull 104 may include two or more discrete projections 132 that each extend around a portion of the perimeter 130. As illustrated in FIG. 6B, the projection 132 is centrally located along the perimeter 130 such that the projection 132 defines a first, outer, recess 136 between the projection 132 and an exterior surface of the hull 104 and a second, inner recess 140 between the projection 132 and an interior surface of the hull 104. It will be appreciated that each of the first and second recesses 136, 140 is therefore configured to receive first and second portions of the cap 108 when the cap 108 is in the first configuration, as will be discussed in greater detail below.

[0024] As best illustrated by FIGS. 7A-7C, the cap 108 has a shape and size that generally matches that shape and size of the hull 104. More particularly, the cap 108 includes a perimeter 144 that has a shape and size that generally matches the shape and size of the perimeter 130 of the hull 104. The cap 108 also includes a channel 148 that is defined by and between outer and inner walls 152, 156 and matches the shape and size of the projection 132 of the hull 104. As such, the channel 148 extends around at least a portion of the perimeter 144 of the cap 108. In this example, the channel 148 extends around an entirety of the perimeter 144. In other examples, however, the channel 148 may only extend around part of the perimeter 144 of the cap 108 or the cap 108 may include two or more discrete channels 148 that each extend around a portion of the perimeter 130.

[0025] As best illustrated by FIGS. 7A-7D and 8, the cap 108 also includes a sealing element 160 disposed in the channel 148. The sealing element 160 preferably takes the form of a bulb gasket made of rubber, as seen in FIGS. 7D and 8, though in other examples, the sealing element 160 can be a different type of seal (e.g., an O-ring, a metal gasket). In any event, the sealing element 152 also extends around at least a portion of the perimeter 144 of the cap 108. Thus, in this example, the sealing element 160 extends around an entirety of the perimeter 144 of the cap 108. In another example, e.g., when the channel 148 only extends around part of the perimeter 144, the sealing element 160 may only extend around part of the perimeter 144. In yet another example, e.g., when the cap 108 includes two or more discrete channels 148, the cap 108 may include two or more sealing elements 160 disposed in the two or more discrete channels 148, respectively.

[0026] It will be appreciated that when the cap 108 is in the first configuration (shown in FIGS. 1-4 and 8), the perimeter 144 of the cap 108 is aligned with the perimeter 130 of the hull 104 such that the projection 132 of the hull 104 is partially disposed in the channel 148 of the cap 108. In turn, the projection 132 of the hull 104 sealingly engages the sealing element 160 disposed in the channel 148 while the outer and inner walls 152, 156 of the cap 108 occupy at least a portion of the outer and inner recesses 136, 140, respectively. Optionally, and in this example, the maritime vehicle 100 also includes an additional seal 164 that is coupled to the maritime vehicle 100 so as to surround an interface between a bottom surface of the cap 108 (in this case the walls 152, 156) and a top surface of the hull 104 (in this case the projection 132) when the cap 108 is in the first configuration. In this example, the additional seal 164 is a Z-shaped rubber seal having one portion secured to the hull

104 and another portion secured to the cap 108. In other examples, however, the additional seal 164 can have a different shape and/or can be made of a different material. In any case, these components help to substantially prevent any water from the body of water and objects in the water (or the ambient environment more generally) from entering the interior 126 of the hull 104 or an interior of the cap 108 via the cap 108 or the interface between the hull 104 and the cap 108.

[0027] Conversely, it will be appreciated that when the cap 108 is in the second configuration, the projection 132 of the hull 104 is at least partially disposed outside of, or spaced from, the channel 148 of the cap 108. In some examples, the entire projection 132 will be disposed outside of the entire channel 148 of the cap 108. In other examples, part of the projection 132 will be disposed within part of the channel 148, respectively, and a remainder of the projection 132 will be disposed outside of a remainder of the channel 148. In any event, when the cap 108 is in the second configuration, the interior 126 of the hull 104 (and the interior of the cap 108) are accessible. It will also be appreciated that in other examples, the cap 108 may include the projection 132 and the hull 104 may include the channel 148 and the sealing element 160, and the cap 108 would still be reconfigurable between the first and second configurations in the manner described herein.

[0028] The cap 108 is generally removably coupled to the hull 104 via a hinge system that facilitates movement of the cap 108 between the first and second configurations described herein and yet helps to sealingly enclose the interior 126 of the hull 104 when the cap 108 is in the first configuration. In this example, and as illustrated in FIGS. 1, 3, 4, and 8, the hinge system takes the form of a plurality of latches 200 disposed around at least a portion (if not the entirety) of a perimeter of the maritime vehicle 100. In this example, the latches 200 are disposed on both sides 120 of the hull 104 and both sides of the cap 108 but not on or along the front and rear sides of the hull 104 or the cap 108, and each of the latches 200 has two ends—a first end 202 and a second end 204 opposite the first end 202—as well as an arm 206 that extends between and connects the first and second ends 202, 204. In this example, the first end 202 of each latch 200 is fixedly coupled (e.g., via one or more fasteners) to an exterior surface of one of the sides 120 of the hull 104, whereas the second end 204 of each latch 200 is fixedly coupled (e.g., via one or more fasteners) to an exterior surface of one of the sides of the cap 104. When the cap 108 is in the first configuration, and when the first and second ends 202, 204 are fixedly coupled to the hull 104 and the cap 108, respectively, the arm 206 of each latch 200 applies a tension force on both the first and second ends 202, 204 so as to help maintain the sealed engagement between the hull 104 and the cap 108 discussed above. At the same time, the first end 202 and/or the second end 204 of one or more latches 200 can be removed from the hull 104 and/or the cap 108, respectively, which allows the cap 108 to be moved from the first configuration to the second configuration. In some cases, some of the latches 200 can be removed from the hull 104 and/or the cap 108 so as to allow the cap 108 to be rotated relative to the hull 104 while the cap 108 is still secured to the hull 104 (via the remaining latches 200). In other cases, each of the latches 200 can be removed from the hull 104 and/or the cap 108 so as to allow the cap 108 to be

completely removed and separated from the hull 104. In either case, the cap 108 is manipulated to allow access to the interior 126 of the hull 104.

[0029] It will be appreciated that in other examples, the hinge system can facilitate the desired movement of the cap 108 relative to the hull 104 without being removed from the hull 104 or the cap 108. For example, the latches 200 can facilitate the desired movement of the cap 108 without being removed from the hull 104 or the cap 108. As another example, the latches 200 may only be positioned along one side 120 of the hull 104 and one side of the cap 108, in which case the other side of the cap 108 is freely movable relative to the other side of the hull 104 via the latches 200. It will also be appreciated that in other examples, the hinge system can be replaced by a locking system that serves to permanently couple the cap 108 to the hull 104 to permanently conceal or cover the components within the interior 126 of the hull 104.

[0030] The maritime vehicle 100 further includes one or more pick points that allow the maritime vehicle 100 to be picked up and carried from one location (e.g., land) to another location (e.g., the body of water) without special tooling. In this example, the maritime vehicle 100 includes a single, load bearing pick point 210 that is coupled to a top, exterior surface 214 of the cap 108. In this example, the single, load bearing pick point 210 is defined by a U-shaped bolt that is coupled to the top, exterior surface 214 of the cap 108. Preferably, the single pick point 210 is located at or immediately adjacent to a center of gravity of the maritime vehicle 100 (when the cap 108 is in the first configuration). In turn, the maritime vehicle 100 can be safely and easily picked up via the single pick point 210 and transported as needed. In other examples, however, the maritime vehicle 100 may include multiple pick points coupled to the cap 108 (e.g., forward and rearward parts of the top, exterior surface 214). In yet other examples, the maritime vehicle 100 may include a single pick point coupled to a portion of the hull 104 and/or may include one or more pick points coupled to the hull 104 and one or more pick points coupled to the cap 108.

[0031] The maritime vehicle 100 also includes a plurality of bulkheads 220 arranged within the hull 104. The bulkheads 220 divide the maritime vehicle 100 into a plurality of different compartments for receiving and retaining different components in the maritime vehicle 100.

[0032] The maritime vehicle 100 also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle 100 as well as data about the environment surrounding the maritime vehicle 100 (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle 100. The sensors can include, for example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle 100, the pressure of the water in which the maritime vehicle 100 is disposed, the pressure within the maritime vehicle 100), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle 100, a temperature of ambient air external to the maritime vehicle 100, a temperature of water in which the maritime vehicle 100 is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors,

compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water sensors (e.g., a float switch, a capacitive sensor, an ultrasonic sensor, an electrical water sensor, etc.) to determine when water is present and/or present to a given extent (e.g., at a certain volume or level), one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof. Further details regarding the sensor system are described in U.S. Provisional Application Nos. 63/698,453, titled “Autonomous Maritime Surface Vehicles” and filed on Sep. 24, 2024, and 63/701,166, titled “Autonomous Maritime Surface Vehicles” and filed on Sep. 30, 2024, the contents of which are each hereby incorporated by reference in its entirety.

[0033] The maritime vehicle **100** also includes a power system that is generally configured to power the maritime vehicle **100** (and the components of the maritime vehicle **100**). The power system generally includes a thrust system and one or more power sources configured to power the thrust system (and the other components within the maritime vehicle **100**). The thrust system is generally configured to propel the maritime vehicle **100** in/on/along the water. The thrust system can be a propeller-based thrust system or can be a jet pump-based thrust system such as the jet pump assembly described in U.S. Provisional Application No. 63/561,166, titled “Jet Pump Assembly for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The one or more power sources can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle **100**, hydrogen stored in hydrogen tanks carried by the maritime vehicle **100**, solar panels (e.g., mounted to an exterior of the vehicle **100**), or other sources. The maritime vehicle **100** illustrated in FIGS. 1-8 includes four battery assemblies each including a rechargeable battery. The maritime vehicle **100** illustrated in FIGS. 1-8 also includes a retention assembly for the four battery assemblies, e.g., the retention assembly described in U.S. Provisional Application No. 63/561,063, titled “Power System for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The maritime vehicle **100** generally also includes a cooling system configured to cool the thrust system and/or the one or more power sources, thereby preventing these components from overheating and leading to failure of the maritime vehicle **100**. For example, the maritime vehicle **100** can include the cooling system described in U.S. Provisional Application No. 63/561,181, titled “Micro-Keel Cooler for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety.

[0034] In operation, the maritime vehicle **100** may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull **104**, attached to the exterior surface of the hull **104**, or attached to the exterior surface of the cap **108** prior to deployment.

Likewise, retrieved payloads can be stored in the hull **104**, attached to and stored on the exterior surface of the hull **104**, or attached to and stored on the exterior surface of the cap **108**.

[0035] The maritime vehicle **100** can also include other systems to help with the operation of the maritime vehicle **100**, for example a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle **100** in the water, regardless of whether the maritime vehicle **100** is stationary or on the move. To this end, the maritime vehicle **100** may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle **100**. Alternatively or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle **100**. The ballast system may also provide for the selective submerging and resurfacing of the maritime vehicle **100** in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle **100** and to guide the maritime vehicle **100** to its desired location in the body of water (or in a different body of water). In some examples, and as illustrated in FIGS. 1-3, the vision system may be sealingly mounted to a front surface of the cap **108**. The vision system is generally configured to capture, process, and analyze images obtained by one or more cameras (e.g., a stereo camera and/or an infrared camera) and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle **100** (including objects in that environment). Further details regarding the vision system are described in U.S. Provisional Application No. 63/561,301, titled “Vision System for Maritime Vehicle,” and filed on Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety.

[0036] The maritime vehicle **100** further includes a communications system that is generally configured to facilitate communication (i) between the maritime vehicle **100** and one or more central (remote) controllers, (ii) between the maritime vehicle **100** and and/or one or more other maritime vehicles **100** and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle **100**. The communications system generally includes one or more local (on-board) controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle **100** and the central controller(s) or other maritime vehicles **100**. For example, the maritime vehicle **100** includes a plurality of antennae **228** disposed on the exterior of the cap **108** as well as a plurality of antennae **232** disposed in the hull **104**.

[0037] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles **100** or military assets) and to perform automated operations of the maritime vehicle **100** based on that data. More particularly, the one or more local controllers are configured to control the maritime vehicle **100** based on that data. In some examples, the maritime vehicle **100** includes a plurality of different local controllers. For example, the maritime vehicle **100** can include one or more thrust con-

trollers (for controlling the operation of the thrust system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0038] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle **100** includes several additional components. For example, the maritime vehicle **100** includes various sealing elements configured to provide seals between different components of the vehicle **100** (or between the vehicle **100** and the environment surrounding the vehicle **100**). As another example, the maritime vehicle **100** also includes various fasteners that help to couple the components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes cabling that helps to communicatively couple components of the maritime vehicle **100** together. Indeed, the maritime vehicle **100** can include electrical cabling conveyed with the cap **108** through the bulkheads **220** or retained with hard wiring. As yet another example, the maritime vehicle **100** includes various electrical components that help to operate the maritime vehicle **100**, e.g., one or more relay boards, one or more DC-DC converters, one or more supervisor boards, and/or one or more brain boards.

[0039] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

1. An autonomous maritime surface vehicle (AMSV), the AMSV comprising:

a hull;

a cap removably coupled to the hull, the cap being reconfigurable between a first configuration, in which the cap is coupled to and sealingly encloses at least a portion of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting access to the interior of the hull.

2. The AMSV of claim **1**, wherein in the first configuration the cap sealingly encloses an entirety of the interior of the hull.

3. The AMSV of claim **1**, further comprising one or more pick points coupled to an exterior surface of the cap.

4. The AMSV of claim **1**, wherein the hull is a mono-hull.

5. The AMSV of claim **1**, further comprising a hinge system that removably couples the cap to the hull.

6. The AMSV of claim **5**, wherein the hinge system includes a plurality of latches disposed around at least a portion of a perimeter of the AMSV.

7. The AMSV of claim **1**, further comprising a plurality of bulkheads arranged within the hull and configured to separate the interior of the hull into separate compartments.

8. The AMSV of claim **1**, wherein when the cap is in the first configuration, a bottom surface of the cap engages a top surface of the hull, further comprising a seal surrounding an interface between the bottom surface of the cap and the top surface of the hull.

9. The AMSV of claim **1**, wherein the cap includes a channel that extends along at least a portion of a perimeter of the cap, the channel sized to receive a top surface of the hull when the cap is in the first configuration.

10. An autonomous maritime surface vehicle (AMSV), the AMSV comprising:

a hull;

a cap removably coupled to the hull, the cap being movable between a first configuration, in which the cap is coupled to and sealingly encloses an entirety of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting access to the interior of the hull; and a single pick point coupled to a top, exterior surface of the cap.

11. The AMSV of claim **10**, wherein the hull is a mono-hull.

12. The AMSV of claim **10**, further comprising a hinge system that removably couples the cap to the hull.

13. The AMSV of claim **12**, wherein the hinge system includes a plurality of latches disposed around at least a portion of a perimeter of the AMSV.

14. The AMSV of claim **10**, further comprising a plurality of bulkheads arranged within the hull and configured to separate the interior of the hull into separate compartments.

15. The AMSV of claim **10**, wherein when the cap is in the first configuration, a bottom surface of the cap engages a top surface of the hull, further comprising a seal surrounding an interface between the bottom surface of the cap and the top surface of the hull.

16. The AMSV of claim **10**, wherein the cap includes a channel that extends along at least a portion of a perimeter of the cap, the channel sized to receive a top surface of the hull when the cap is in the first configuration.

17. An autonomous maritime surface vehicle (AMSV), the AMSV comprising:

a hull, the hull including a projection that extends around at least a portion of a perimeter of the hull;

a cap removably coupled to the hull, the cap including a channel that extends along at least a portion of the perimeter of the hull;

a sealing element disposed in the channel of the cap;

wherein the cap is movable between a first configuration, in which the projection of the hull is disposed in the channel of the cap and sealingly engages the sealing element, thereby sealingly enclosing at least a portion of an interior of the hull, and a second configuration, in which the projection of the hull is at least partially spaced from the channel of the cap, thereby permitting access to the interior of the hull.

18. The AMSV of claim **17**, wherein when the cap is in the first configuration, further comprising a seal surrounding an interface between the projection of the hull and channel of the cap.

19. The AMSV of claim **17**, further comprising a hinge system that removably couples the cap to the hull, wherein

the hinge system includes a plurality of latches disposed around at least a portion of a perimeter of the AMSV.

20. The AMSV of claim **17**, further comprising a vision system sealingly mounted to a front surface of the cap, wherein the vision system comprises a stereo camera and an infrared camera.

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